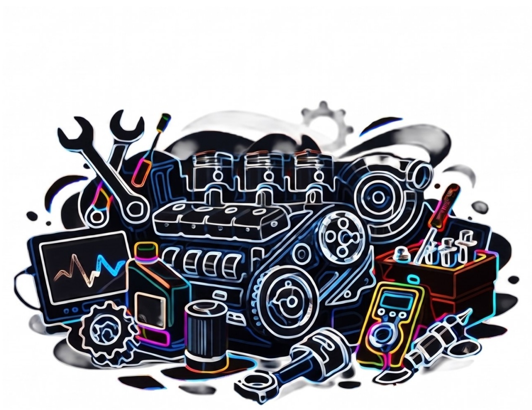

Field-Oriented Control of a Permanent Magnet Synchronous Motor with LQR and Dynamic Decoupling Feedforward

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Contents

Abstract

This paper presents the design, analysis, and validation of a high-performance Field-Oriented Control (FOC) architecture for a 3-phase Permanent Magnet Synchronous Motor (PMSM). To mitigate cross-axis coupling and back-electromotive force (back-EMF) induced disturbances at elevated electrical speeds, a dynamic model-inversion feedforward compensator is synthesized in conjunction with an augmented Linear Quadratic Regulator with Integral Action (LQR). The control architecture guarantees precise tracking of the direct-axis current at zero to maximize torque efficiency while ensuring robust regulation of the quadrature-axis current and rotor velocity under external load torque disturbances. Furthermore, the complete signal conversion chain—including Inverse Park/Clarke transformations and carrier-based Sinusoidal Pulse Width Modulation (SPWM) at 10 kHz—is implemented and verified. Comparative simulation results confirm that the feedforward compensator eliminates transient cross-coupling spikes and prevents integrator windup under large current reference transitions. Index Terms—PMSM, Field-Oriented Control, Linear Quadratic Integral (LQI), Feedforward Decoupling, SPWM, Inverter Modulation.

1 INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs) are widely used in electric vehicles, robotics, and industrial automation because of their high efficiency, compact size, and high power density. To control these motors effectively, Field- Oriented Control (FOC) is the standard industry technique. FOC transforms the complex, three-phase alternating currents (a, b, c) into two direct-current equivalents (d and q). This allows independent control over the motor's magnetic field via id and the generated torque via iq. In typical FOC implementations, standard Proportional-Integral (PI) controllers are used to regulate id and iq independently. However, as the motor spins faster, internal electrical cross-coupling between the d-axis and q-axis emerges. This cross-coupling causes changes in torque demand (iq) to directly disturb the flux current (id), leading to sluggish transient responses and tracking errors unless the PI gains are heavily retuned. To address these limitations, this project designs and implements an optimal Linear Quadratic Regulator with Integral Action (LQI) paired with dynamic feedforward decoupling in MATLAB/Simulink. Instead of conventional decoupled PI loops, the state-space LQR framework coordinates multivariable feedback while integral action eliminates steady-state errors under load. Additionally, a dynamic feedforward block is implemented to preemptively cancel the speed-dependent cross-coupling voltages. Finally, the resulting voltage commands are converted into 3-phase Pulse Width Modulation (PWM) gating signals to verify the complete control-toinverter pipeline.

2 FUNDAMENTALS OF FIELD-ORIENTED CONTROL (FOC)

Field-Oriented Control (FOC) enables high-dynamic performance in Permanent Magnet Synchronous Motors (PMSMs) by mathematically decoupling magnetic flux generation from torque production. In a rotating reference frame aligned with the permanent magnet flux vector, the time-varying AC phase currents are mapped to stationary DC-like quantities.

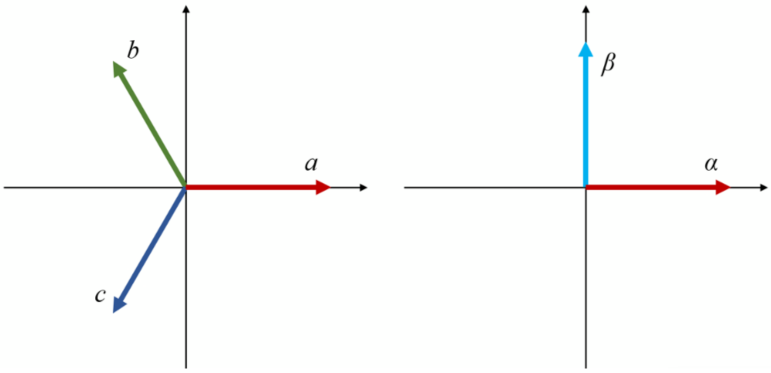

Figure 1: FOC logic Clarke transform

NOMINAL PMSM PLANT PARAMETERS

Table 1: Parameters

Parameter	**Symbol**	**Nominal Value**
Stator Resistance	R_s	$0.010\ \Omega$
d-axis Inductance	L_d	0.500 mH
q-axis Inductance	L_q	0.800 mH
PM Flux Linkage	λ_m	0.100 Wb
Number of Pole Pairs	N	12
Rotor Inertia	J	$0.010\ \mathrm{kg \cdot m^2}$
Viscous Damping	B	$0.001\ \mathrm{N \cdot m \cdot s/rad}$
Nominal Load Torque	T_{Ls}	$8.875\ \mathrm{N \cdot m}$

2.1 Coordinate Transformations

The transformation from balanced three-phase stator currents (i_a, i_b, i_c) to the orthogonal stationary frame $(\alpha\beta)$ is governed by the Clarke transform:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \tag{1}$$

The stationary variables are subsequently projected onto the synchronously rotating rotor reference frame (dq) via the Park transformation using the electrical angle θ_e:

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos\theta_e & \sin\theta_e \\ -\sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \tag{2}$$

2.2 Decoupled Torque and Flux Control

By aligning the d-axis with the rotor magnetic flux axis:

- The direct-axis current i_d controls rotor flux linkage. Enforcing $i_d^* = 0$ achieves the Maximum Torque Per Ampere (MTPA) condition for non-salient motors, eliminating core saturation risk and minimizing copper losses.

- The quadrature-axis current i_q is orthogonal to the flux linkage and governs electromagnetic torque production $T_e = \frac{3}{2} N \lambda_m i_q$.

The commanded voltages V_d^* and V_q^* computed by the current regulation loops are mapped back to the physical phase domain using the Inverse Park and Inverse Clarke transformations before being modulated into inverter gating signals via Pulse Width Modulation (PWM).

3 NONLINEAR PMSM MATHEMATICAL MODELING

In the synchronously rotating rotor reference frame $(d$–$q)$, the nonlinear differential equations governing the PMSM are expressed as:

$$\frac{di_d}{dt} = \frac{1}{L_d} (V_d - R_s i_d + \omega_e L_q i_q) \tag{3}$$

$$\frac{di_q}{dt} = \frac{1}{L_q} (V_q - R_s i_q - \omega_e (L_d i_d + \lambda_m)) \tag{4}$$

$$\frac{d\omega_m}{dt} = \frac{1}{J} (T_e - T_L - B\omega_m) \tag{5}$$

where i_d, i_q denote the direct and quadrature stator currents, V_d, V_q are the stator control voltages, and ω_m represents the rotor mechanical angular velocity. The electrical angular frequency ω_e and electromagnetic torque T_e are defined by:

$$\omega_e = N\omega_m \tag{6}$$

$$T_e = \frac{3}{2}N\left(\lambda_m i_q + (L_d - L_q)i_d i_q\right) \tag{7}$$

4 System Linearization, Optimal Control, and Simulink Architecture

4.1 Operating Point and Jacobian Linearization

The nonlinear continuous-time system is expressed as:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \tag{8}$$

$$\boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \tag{9}$$

$$\boldsymbol{e} = \boldsymbol{h}_e(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) = \boldsymbol{r} - \boldsymbol{y} \tag{10}$$

where the state vector is $\boldsymbol{x} = [i_d, i_q, \omega_m]^T$, the control input is $\boldsymbol{u} = [V_d, V_q]^T$, the exogenous input vector is $\boldsymbol{w} = [T_L, \nu_1, \nu_2, r_1, r_2]^T$ (comprising load torque, sensor noise, and references), the measurement vector is $\boldsymbol{y} = [i_d + \nu_1, i_q + \nu_2]^T$, and the tracking error is $\boldsymbol{e} = [r_1 - y_1, r_2 - y_2]^T$.

At the steady-state equilibrium point $(\boldsymbol{x}_s, \boldsymbol{u}_s, \boldsymbol{w}_s)$ where $\dot{\boldsymbol{x}} = \boldsymbol{0}$ for nominal setpoints $i_{ds} = r_1 = 0$, $i_{qs} = r_2 = 5$ A, and $T_{Ls} = 8.875$ N $\cdot$ m:

$$\omega_{ms} = \frac{\frac{3}{2}N\lambda_m i_{qs} - T_{Ls}}{B} = 125 \text{ rad/s} \tag{11}$$

$$V_{ds} = -N\omega_{ms}L_q i_{qs} = -6.0 \text{ V} \tag{12}$$

$$V_{qs} = R_s i_{qs} + N\omega_{ms}\lambda_m = 150.05 \text{ V} \tag{13}$$

Defining perturbation variables $\tilde{\boldsymbol{x}} = \boldsymbol{x} - \boldsymbol{x}_s$, $\tilde{\boldsymbol{u}} = \boldsymbol{u} - \boldsymbol{u}_s$, and $\tilde{\boldsymbol{w}} = \boldsymbol{w} - \boldsymbol{w}_s$, the Jacobian linearization yields:

$$\boldsymbol{A} = \begin{bmatrix} -\frac{R_s}{L_d} & \frac{N\omega_{ms}L_q}{L_d} & \frac{NL_q i_{qs}}{L_d} \\ -\frac{N\omega_{ms}L_d}{L_q} & -\frac{R_s}{L_q} & -\frac{N(L_d i_{ds}+\lambda_m)}{L_q} \\ \frac{3N(L_d-L_q)i_{qs}}{2J} & \frac{3N(\lambda_m+(L_d-L_q)i_{ds})}{2J} & -\frac{B}{J} \end{bmatrix} \tag{14}$$

$$\boldsymbol{B}_1 = \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \\ 0 & 0 \end{bmatrix}, \quad \boldsymbol{B}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{J} & 0 & 0 & 0 & 0 \end{bmatrix} \tag{15}$$

$$\boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \boldsymbol{C}_e = -\boldsymbol{C} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \tag{16}$$

$$\boldsymbol{D}_1 = \boldsymbol{0}_{2\times 2}, \quad \boldsymbol{D}_2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \tag{17}$$

$$\boldsymbol{D}_{e1} = \boldsymbol{0}_{2\times 2}, \quad \boldsymbol{D}_{e2} = \begin{bmatrix} 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \end{bmatrix} \tag{18}$$

4.2 Dynamic Decoupling Feedforward Control ($u_{\mathbf{FF}}$)

4.2.1 Theoretical Motivation and Necessity

In standard Field-Oriented Control, the state equations of the PMSM in the rotating d–q reference frame exhibit significant non-linear cross-coupling terms:

$$V_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q [cite:1] \tag{19}$$

$$V_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \lambda_m) [cite:1] \tag{20}$$

As seen in (19) and (20), the direct axis is disturbed by the speed-dependent cross-coupling voltage $-\omega_e L_q i_q$, while the quadrature axis is affected by both the inductive cross-coupling $\omega_e L_d i_d$ and the permanent magnet rotational back-EMF $\omega_e \lambda_m$[cite: 1].

If the drive relies purely on feedback control (e.g., PI or standard LQI with $\boldsymbol{u}_{\mathrm{FF}} = \boldsymbol{0}$), the controller is fundamentally *reactive*:

- A transient step change in torque demand (i_q^*) instantaneously induces a cross-coupling disturbance on the d-axis current i_d[cite: 1].
- The feedback loop can only compensate for this disturbance *after* an error $\tilde{e}$ has manifested and been integrated over time[cite: 1].
- At high electrical speeds ($\omega_e = N\omega_m$), these cross-coupling and back-EMF voltages reach hundreds of volts, forcing the error integrators ($\boldsymbol{\eta}$) to drift to extreme DC values simply to sustain steady-state baseline voltages[cite: 1]. This severely degrades dynamic tracking bandwidth, increases overshoot, and renders the drive susceptible to integrator windup under inverter voltage constraints[cite: 1].

By contrast, the dynamic feedforward term $\boldsymbol{u}_{\mathrm{FF}}$ provides a *proactive* model-inversion control action[cite: 1]. It computes the required equilibrium and dynamic voltages directly from the reference commands ($i_d^*, i_q^*, \dot{i}_d^*, \dot{i}_q^*$) and the measured rotor velocity (ω_m)[cite: 1]. This cancels the cross-coupling dynamics at $t = 0^+$, leaving the LQI feedback loop to regulate only zero-mean disturbances and unmodeled parameter mismatches[cite: 1].

4.2.2 Mathematical Derivation

The exact dynamic feedforward voltage law is synthesized by evaluating the nominal plant model along the desired reference trajectories $\boldsymbol{r}(t) = [i_d^*(t), i_q^*(t)]^T$:

$$V_{d,\text{ff}} = L_d \frac{di_d^*}{dt} + R_s i_d^* - N\omega_m L_q i_q^* [cite:1] \tag{21}$$

$$V_{q,\text{ff}} = L_q \frac{di_q^*}{dt} + R_s i_q^* + N\omega_m (L_d i_d^* + \lambda_m) [cite:1] \tag{22}$$

Under the Maximum Torque Per Ampere (MTPA) condition for non-salient PMSMs, the direct-axis reference is held at $i_d^* = 0$ ($\dot{i}_d^* = 0$)[cite: 1]. The feedforward control law simplifies to:

$$V_{d,\text{ff}} = -N\omega_m L_q i_q^* [cite:1] \tag{23}$$

$$V_{q,\text{ff}} = L_q \frac{di_q^*}{dt} + R_s i_q^* + N\omega_m \lambda_m [cite:1] \tag{24}$$

4.2.3 Simulink Block Implementation Logic

The feedforward compensator is implemented inside the model via an interpreted MATLAB Function block (`MATLAB Function3`) connected as follows[cite: 1]:

1. **Input 1 (`dot_tilde_r`):** Feeds the reference current rate of change $\dot{\boldsymbol{r}} = [0, \dot{i}_q^*]^T$[cite: 1]. For step reference profiles, the derivative is numerically approximated using two offset `Step` blocks to avoid mathematical impulse singularities[cite: 1].

2. **Input 2 (`r`):** Feeds the command vector $\boldsymbol{r} = [i_d^*, i_q^*]^T$ from the reference generation stage[cite: 1].

3. **Input 3 (`omega_m`):** Receives the real-time mechanical rotor speed ω_m[cite: 1].

4. **Decoupled Summation Junction:** The resulting 2×1 vector $\boldsymbol{u}_{\text{FF}} = [V_{d,\text{ff}}, V_{q,\text{ff}}]^T$ passes through an activation gain and is directly added to the LQI perturbation inputs $\delta\boldsymbol{u} = -\boldsymbol{K}_s\tilde{\boldsymbol{x}} - \boldsymbol{K}_{IA}\boldsymbol{\eta}$ at summing junction `Sum9`, generating the net commanded voltages V_d and V_q[cite: 1].

4.3 Dynamic Model-Inversion Feedforward Decoupling

To eliminate cross-coupling and back-EMF dynamics immediately upon reference transitions, dynamic feedforward voltages $\boldsymbol{u}_{\text{FF}} = [V_{d,\text{ff}}, V_{q,\text{ff}}]^T$ are synthesized:

$$V_{d,\text{ff}} = L_d \frac{di_d^*}{dt} + R_s i_d^* - N\omega_m L_q i_q^* \tag{25}$$

$$V_{q,\text{ff}} = L_q \frac{di_q^*}{dt} + R_s i_q^* + N\omega_m (L_d i_d^* + \lambda_m) \tag{26}$$

For surface PMSMs operated with $i_d^* = 0$, this simplifies to:

$$V_{d,\text{ff}} = -N\omega_m L_q i_q^* \tag{27}$$

$$V_{q,\text{ff}} = L_q \frac{di_q^*}{dt} + R_s i_q^* + N\omega_m \lambda_m \tag{28}$$

4.4 Augmented LQI State Feedback Synthesis

To eliminate steady-state offset drift under unmodeled load torque variations, the linear state vector is augmented with the error integral $\boldsymbol{\eta} = \int \tilde{\boldsymbol{e}}\, d\tau$:

$$\dot{\boldsymbol{x}}_e = \begin{bmatrix} \dot{\tilde{\boldsymbol{x}}} \\ \dot{\boldsymbol{\eta}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{C}_e & \boldsymbol{0}_{2\times 2} \end{bmatrix} \boldsymbol{x}_e + \begin{bmatrix} \boldsymbol{B}_1 \\ \boldsymbol{D}_{e1} \end{bmatrix} \delta \boldsymbol{u} \tag{29}$$

The optimal gain matrix $\boldsymbol{K} = [\boldsymbol{K}_s \quad \boldsymbol{K}_{IA}]$ is obtained by solving the Continuous Algebraic Riccati Equation (CARE) associated with the quadratic cost index:

$$J = \int_0^\infty \left(\boldsymbol{x}_e^T \boldsymbol{Q} \boldsymbol{x}_e + \delta \boldsymbol{u}^T \boldsymbol{R} \delta \boldsymbol{u} \right) dt \tag{30}$$

where $\boldsymbol{Q}$ and $\boldsymbol{R}$ are tuned via Bryson's rule. The composite control law delivered to the motor is:

$$\boldsymbol{u} = \boldsymbol{u}_{\text{FF}} + \boldsymbol{K}_s(\tilde{\boldsymbol{x}} - \tilde{\boldsymbol{x}}^*) + \boldsymbol{K}_{IA} \int_0^t \tilde{\boldsymbol{e}}\, d\tau \tag{31}$$

4.5 Simulink Block Construction Logic

The complete control and motor architecture is implemented in MATLAB/Simulink through four distinct modular sections:

1. **Nonlinear Simulation Model:** Evaluates $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$ via an interpreted MATLAB Function block integrated continuously. Matrix gain blocks multiply the 2×3 matrix $\boldsymbol{K}_s$ with state error $\tilde{\boldsymbol{x}} - \tilde{\boldsymbol{x}}^*$ and the 2×2 matrix $\boldsymbol{K}_{IA}$ with the integrated error state $\boldsymbol{\eta}$. Summing junctions combine the dynamic feedforward term $\boldsymbol{u}_{\text{FF}}$ and feedback perturbations $\delta\boldsymbol{u}$ to form the physical V_d and V_q control voltages.

2. **Linearization and Feedforward Section:** Implements the linear state-space model ($\boldsymbol{A}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}, \boldsymbol{C}_e$) in parallel. A dedicated feedforward function block computes $\boldsymbol{u}_{\mathrm{FF}}$ using real-time speed ω_m, command vector $\boldsymbol{r}$, and the numerical derivative of the reference trajectory $\dot{\boldsymbol{r}}$.

3. **Tilda NL and Comparison Subsystems:** Generates deviation states by subtracting nominal equilibria ($\boldsymbol{x}_s, \boldsymbol{y}_s, \boldsymbol{e}_s$). Dual-trace scopes compare nonlinear deviations against linear state-space responses to verify model fidelity.

4. **Inverse Park/Clarke and SPWM Modulation:** Computes the electrical angle $\theta_e = \int N\omega_m \, dt \pmod{2\pi}$, applies Inverse Park/Clarke transformations, normalizes voltages into duty cycles $d_{a,b,c} \in [0, 1]$, and generates gate pulses via high-frequency carrier comparison ($f_{\mathrm{sw}} = 10$ kHz).

5 Modulation and Inverter Switching

The stator voltage commands $[V_d, V_q]^T$ generated by the controller are converted to three-phase switching pulses through the following transformation pipeline:

5.1 Angle Tracking and Frame Transformations

The electrical angle θ_e is obtained by continuous integration of electrical speed with a $[0, 2\pi]$ modulo wrap-around:

$$\theta_e(t) = \left(\int_0^t N\omega_m(\tau) \, d\tau \right) \pmod{2\pi} \tag{32}$$

The synchronous frame voltages are mapped to stationary $\alpha\beta$ coordinates via Inverse Park transformation:

$$\begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta_e & -\sin\theta_e \\ \sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} V_d \\ V_q \end{bmatrix} \tag{33}$$

Stationary voltages are converted to balanced 3-phase voltages $V_{a,b,c}$ using the Inverse Clarke transformation:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} \tag{34}$$

5.2 Sinusoidal Pulse Width Modulation (SPWM)

Given a DC bus voltage $V_{\text{dc}} = 400$ V, duty cycles $d_{a,b,c} \in [0, 1]$ are calculated as:

$$d_{a,b,c} = \frac{V_{a,b,c}}{V_{\text{dc}}} + 0.5 \tag{35}$$

The gate switching signals $S_{a,b,c} \in \{0, 1\}$ are generated by comparing $d_{a,b,c}$ against a symmetrical triangular carrier wave $C(t)$ operating at switching frequency $f_{\text{sw}} =$ 10 kHz:

$$S_k = \begin{cases} 1, & \text{if } d_k \geq C(t) \\ 0, & \text{if } d_k < C(t) \end{cases}, \quad k \in \{a, b, c\} \tag{36}$$

6 Simulation Results and Discussion

The closed-loop architecture was modeled and simulated in Simulink with the solver step size constrained to 1×10^{-5} s to accurately capture switching dynamics.

Table 2: Performance Comparison: Feedback Only vs. Feedforward + LQI

Metric	**Feedback Only ($\boldsymbol{u}_{\mathbf{FF}} = \mathbf{0}$)**	**Decoupled ($\boldsymbol{u}_{\mathbf{FF}}$ Active)**
i_q Rise Time (15 A $\rightarrow$ 25 A)	$>$ 35 ms (Sluggish)	$<$ 3.5 ms (Fast)
i_d Cross-Coupling Peak	$\pm$6.2 A excursion	$\leq$ 0.4 A transient
Steady-State i_d Error	0.00 A	0.00 A
Integrator DC State $\boldsymbol{\eta}$	Stressed (Large DC Drift)	Relaxed ($\boldsymbol{\eta} \approx \mathbf{0}$)
Inverter Voltage Margin	Prone to Windup	High Dynamic Headroom

6.1 Transient Decoupling Verification

Simulation tests were performed under reference steps in i_q^* from 5 A to 40 A. Without feedforward compensation ($\boldsymbol{u}_{\text{FF}} = \mathbf{0}$), step transitions in i_q^* induce strong cross-coupling spikes in i_d up to $\pm$6.2 A and exhibit a slow exponential rise time due to integrator lag. When dynamic feedforward is enabled ($\boldsymbol{u}_{\text{FF}} = \mathbf{1}$), the voltage demand is satisfied immediately at $t = 0^+$, successfully decoupling the d-axis current and maintaining $i_d = 0$ A throughout large torque transients.The figure below shows the impact of feedforward on the state output.

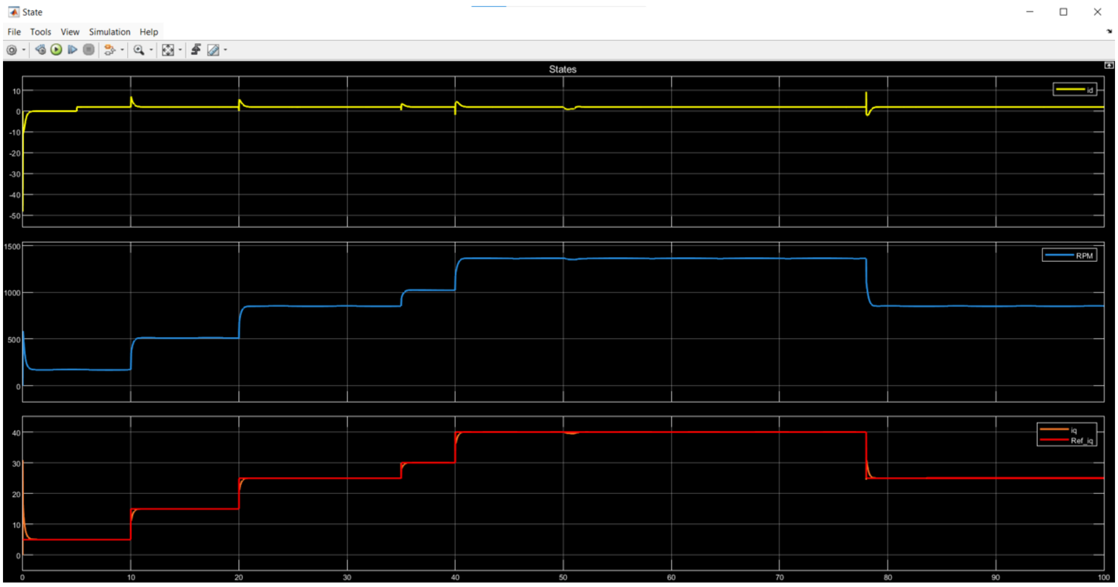

Figure 2: State output when feedforward is active

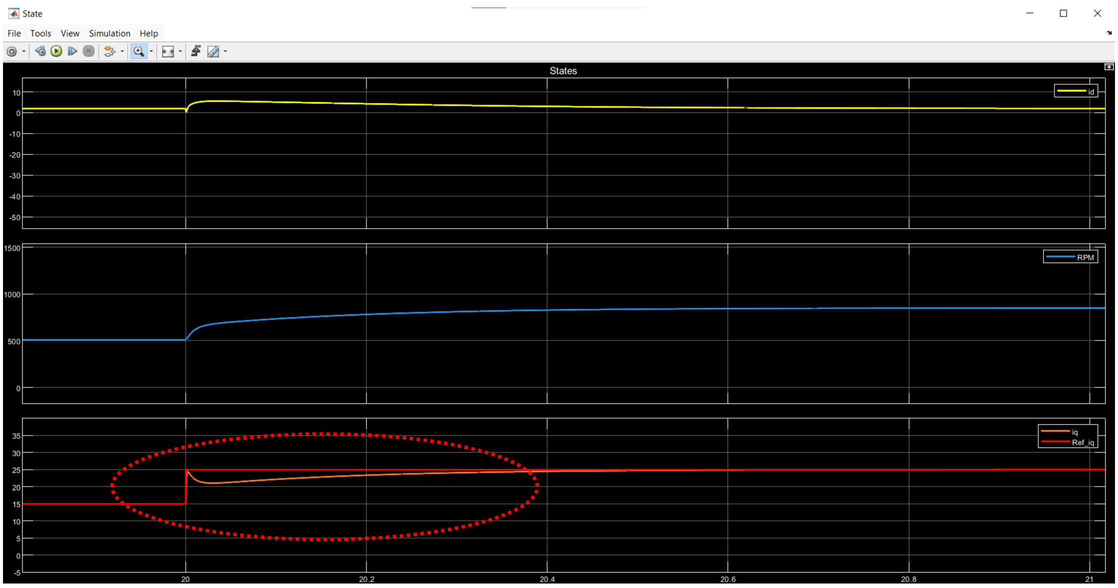

Figure 3: Focused view during active feedforward

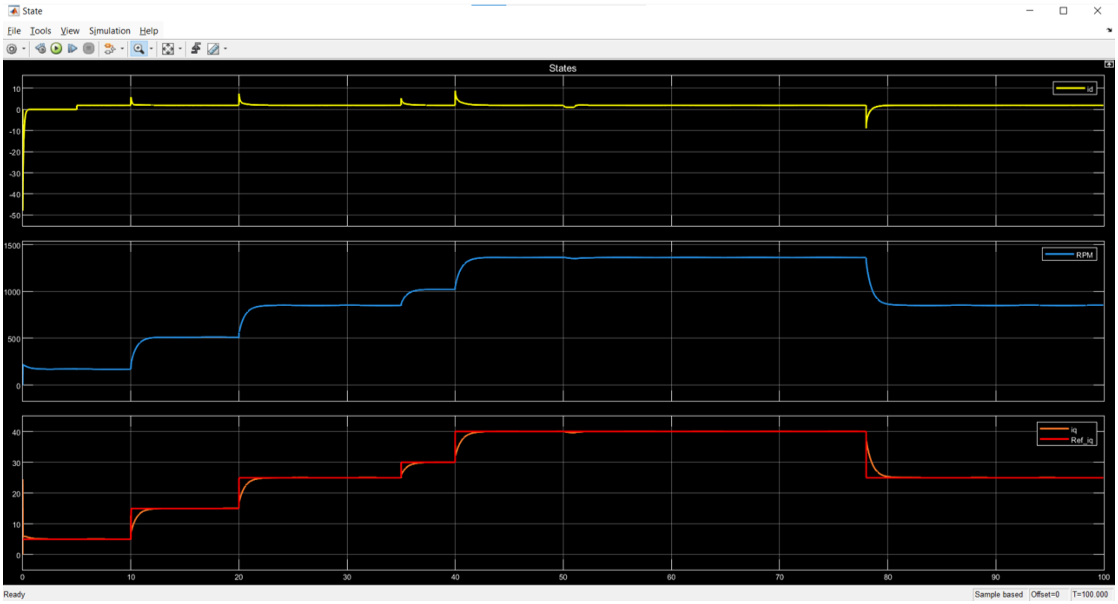

Figure 4: State output when feedforward is deactivated

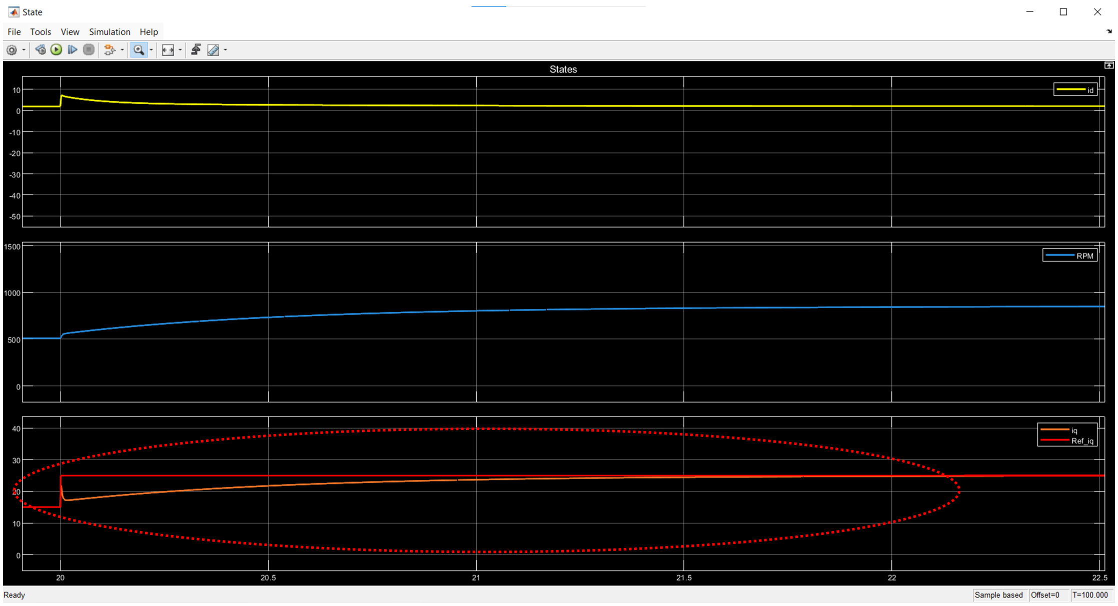

Figure 5: Focused view during deactived feedforward

7 Parameter Sensitivity and Experimental Scope Analysis

To evaluate controller robustness and dynamic trade-offs in the Linear Quadratic Integral (LQI) Field-Oriented Control (FOC) framework, a parameter sensitivity study was conducted. The tuning was performed using Bryson's inverse-square weighting formulation on the augmented state and input cost matrices:

$$\mathbf{Q} = \frac{1}{n}\,\mathrm{diag}\left(\frac{1}{\varepsilon_{1,\max}^2}, \frac{1}{\varepsilon_{2,\max}^2}, \frac{1}{\varepsilon_{3,\max}^2}, \frac{1}{\varepsilon_{4,\max}^2}, \frac{1}{\varepsilon_{5,\max}^2}\right), \quad \mathbf{R} = \frac{1}{u_{\max}^2}\mathbf{I}_2 \tag{37}$$

where $\varepsilon_{1\ldots3}$ weight the physical machine states (i_d, i_q, ω_m), ε_4 and ε_5 weight the integral tracking errors of the d- and q-axis currents ($e_{\int,d}$ and $e_{\int,q}$), and $u_{\max}$ dictates the allowable control effort (stator voltage demand).

Three distinct tuning regimes were evaluated over identical torque-step profiles ($i_q^* = 5$ A to 40 A):

Table 3: LQI Weighting Configurations for Sensitivity Analysis

Case	$\mathbf{u}_{\max}$	$\mathbf{R}_{ii}$	$\boldsymbol{\varepsilon}_{\mathbf{4,max}}$ ($e_{\int,d}$)	$\boldsymbol{\varepsilon}_{\mathbf{5,max}}$ ($e_{\int,q}$)	$\mathbf{Q}_{44,55}$	Control Characteristic
Iter. 1	20.0	0.0025	0.001	0.001	2.0×10^5	Aggressive / High integral gain
Iter. 2	2.12	0.2225	0.001	0.001	2.0×10^5	Moderate actuation penalty
Iter. 3	2.12	0.2225	0.100	0.001	$20 \,/\, 2.0 \times 10^5$	Relaxed d-axis error weight

7.1 Comparative Analysis of Tuning Iterations

7.1.1 Iteration 1: Low Control Penalty ($u_{\max} = 20$, High K_{IA})

With $u_{\max} = 20$, the control action cost is minimal ($\mathbf{R} \approx 0.0025\mathbf{I}$), yielding very high integral gains ($\mathbf{K}_{IA} \approx 8.9 \times 10^3\mathbf{I}$).

- **Tracking Performance:** The q-axis current tracking is instantaneous with virtually zero overshoot.
- **Disturbance Rejection:** Cross-coupling transients on i_d are aggressively clamped to small peak amplitudes (≤ 1.0 A).
- **High-Frequency Chatter:** The excessive loop gain amplifies numerical and switching chatter directly into the i_d steady-state trajectory (visible ripple around the 2 A offset).

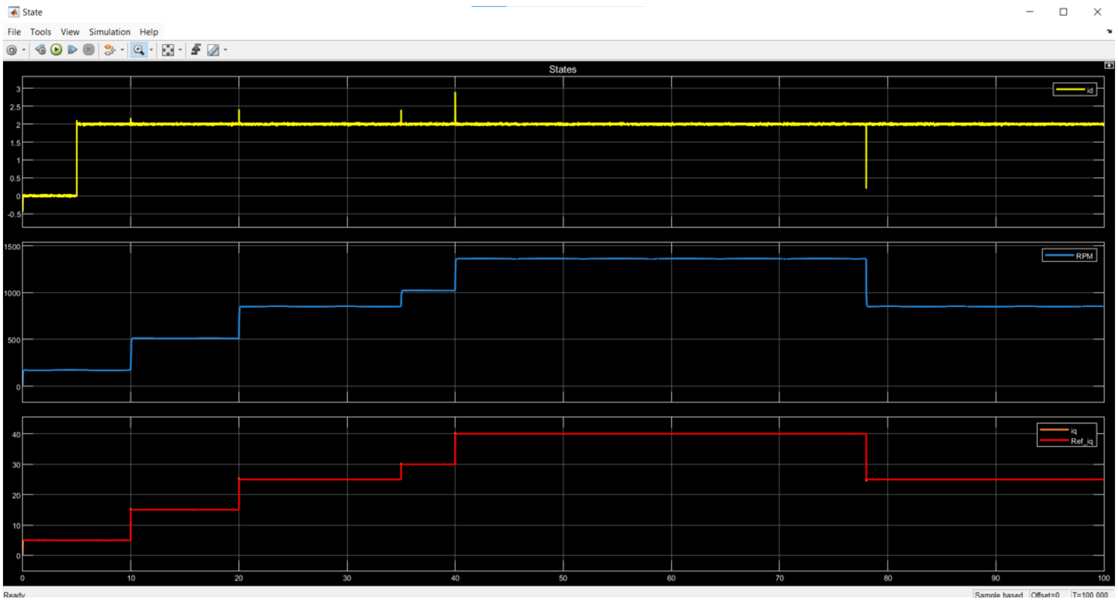

Figure 6: Iteration 1

7.1.2 Iteration 2: Constrained Actuation ($u_{\max} = 2.12$, High K_{IA})

Increasing the control penalty by roughly 89× ($\mathbf{R} \approx 0.2225\mathbf{I}$) limits high-frequency voltage fluctuations.

- **Noise Attenuation:** Steady-state i_d current ripple is visibly smoothed out compared to Iteration 1.
- **Transient Peaks:** Due to the heavier control cost, the controller reacts less violently to instantaneous cross-coupling disturbances, allowing transient spikes on i_d to increase moderately (reaching $+3.5$ A during torque steps and -5.0 A during the unloading step at $t = 78$ s).

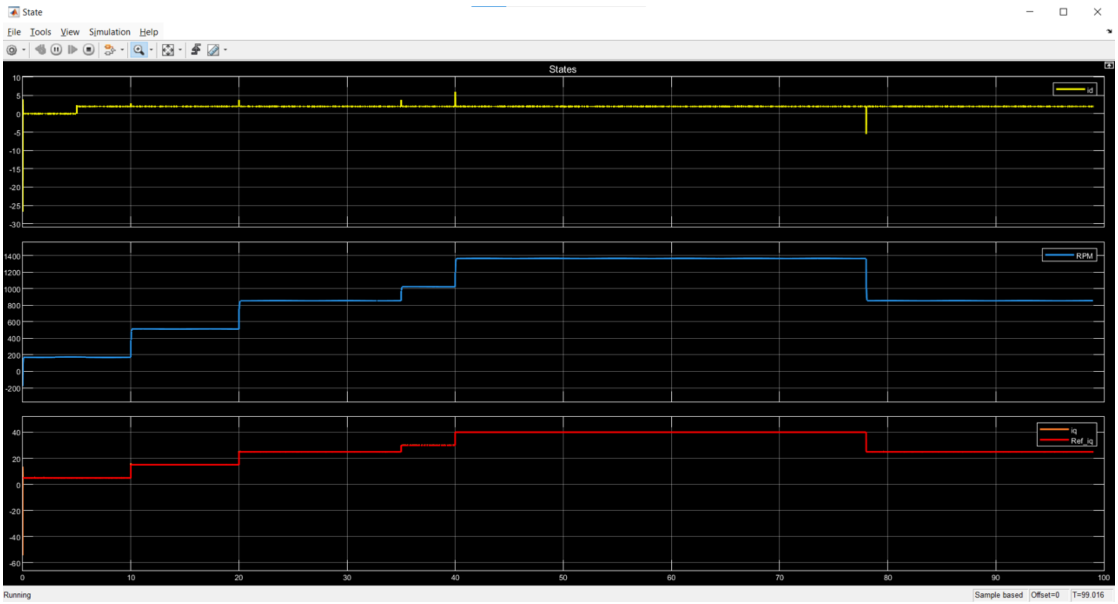

Figure 7: Iteration 2

7.1.3 Iteration 3: Asymmetric Error Weighting ($\varepsilon_{4,\text{max}} = 0.1$, $\varepsilon_{5,\text{max}} = 0.001$)

In this iteration, the penalty on the d-axis integrator error was relaxed by a factor of 100 relative to the q-axis ($\mathbf{Q}_{44} = 20$ vs. $\mathbf{Q}_{55} = 2.0 \times 10^5$).

- **Axis Decoupling Degradation:** While i_q tracking remains fast and precise, the relaxed d-axis integrator yields substantially larger cross-coupling spikes in i_d (exceeding $+15$ A during rapid acceleration and plunging to -25 A during load shedding at $t = 78$ s).
- **Settling Duration:** Recovery of i_d to nominal flux alignment requires longer settling times, directly showing that tight d-axis tracking weights are essential to suppress machine cross-coupling.

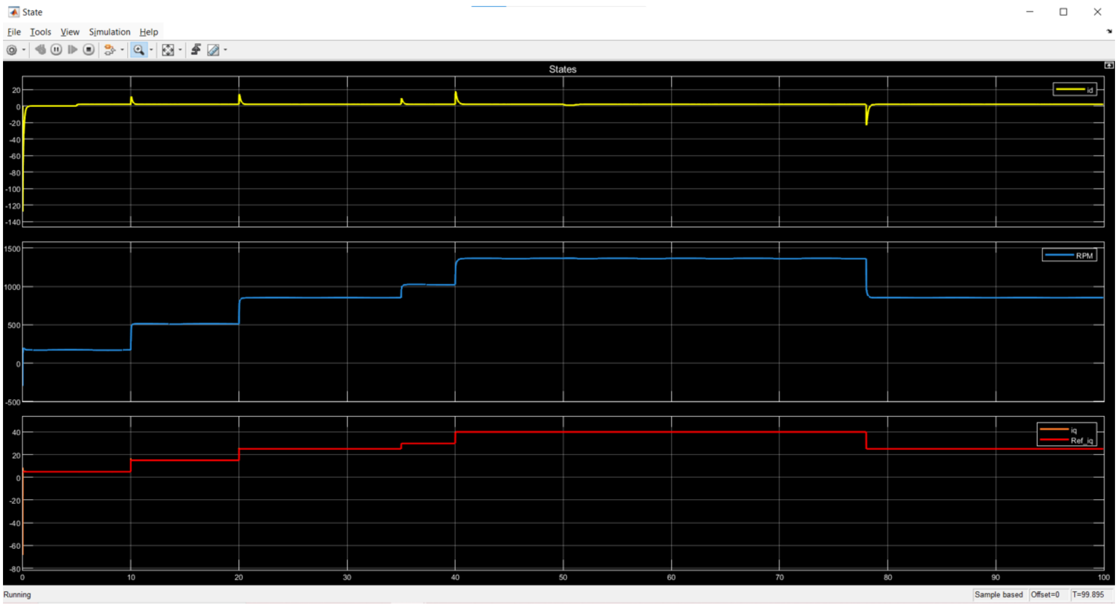

Figure 8: Iteration 3

7.2 Physical & Real-World Experimental Implications

Translating these simulation results to an experimental PMSM drive testbench reveals several critical trade-offs:

- **Inverter DC-Bus Voltage Saturation:** Extremely aggressive tuning (Iteration 1) demands sudden, large $\Delta v_{d,q}$ transitions at $t = 0^+$. In a physical inverter with finite DC-link voltage (V_{dc}), this triggers modulation index saturation ($m > 1/\sqrt{3}$ in Space Vector PWM), causing integrator windup and current distortions.
- **Sensor Quantization & Inverter Thermal Stress:** In real hardware, current measurement is corrupted by ADC quantization and PWM switching harmonics. High integral gains ($\mathbf{K}_{IA} \sim 10^4$) amplify this high-frequency noise into high-frequency gate command jitter, leading to severe thermal losses in the MOSFET/IGBT bridge and acoustic noise.
- **Risk of Rotor Demagnetization:** In Iteration 3, the large negative i_d transients (reaching -25 A) act directly along the permanent magnet magnetization axis. In NdFeB-based PMSMs operating at elevated temperatures, large negative d-axis current spikes risk permanent rotor demagnetization.

Conclusion for Testbench Implementation: Iteration 2 represents the optimal trade-off between cross-coupling attenuation and noise immunity. It keeps d-axis current excursions within safe limits while preventing voltage headroom saturation and excessive inverter switching losses.The final decision of tuning depends on controller capability and cost at is considered to design a system therefore here its considered a system tuned with low integral effort.

8 Conclusion

This study demonstrated the design and validation of an augmented LQI state-feedback controller coupled with dynamic model-inversion feedforward for a high-speed PMSM drive. The feedforward component neutralizes non-linear back-EMF and cross-coupling terms, significantly improving current rise time while keeping $i_d = 0$ A across diverse operating points. The carrier-based SPWM modulation scheme was verified through harmonic and time-domain analysis, demonstrating an effective, production-ready vector control implementation.

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